2026年4月15日

王灏

博士生在读（预计2026年7月毕业）
瑞典查尔姆斯理工大学机械工程学院
+86 18810560620（中国）/ +46(0)731559494（瑞典）
haowang7308@gmail.com / haowang@chalmers.se
Division of Production Systems, Department of Mechanical Engineering, Chalmers University of Technology, 412 96 Gothenburg, Sweden

教育背景

博士生，非刚性物体装配自动化中的计算机视觉技术，查尔姆斯理工大学，哥德堡，瑞典	2021.08 - 2026.07
理学硕士，信息学，爱丁堡大学，爱丁堡，英国	2018.09 - 2019.11
工学学士，通信工程，北京邮电大学，北京，中国	2014.09 - 2018.06

研究经历

博士研究生，生产系统系，机械工程学院，查尔姆斯理工大学，哥德堡，瑞典	2026.01 - 2026.07
博士研究生，生产系统系，工业与材料科学学院，查尔姆斯理工大学，哥德堡，瑞典	2021.08 - 2025.12
研究助理，模式识别国家重点实验室，中国科学院自动化研究所，北京，中国	2019.10 - 2021.06
研究助理，下一代互联网研究中心，北京邮电大学，北京，中国	2017.08 - 2017.10

发表成果

Google Scholar（h-index 9，2026年4月）：https://scholar.google.com/citations?user=hKFj_QwAAAAJ
Scopus（h-index 7，2026年4月）：https://www.scopus.com/authid/detail.uri?authorId=56181425300

期刊

1. **Wang, H.**,Urbanos Uriel, G., El-Nahass, K., Ekered, S., Johansson, B. (2026). Accelerating industrial vision: Systematic robot-assisted dataset preparation for object detection and pose estimation. *Engineering Applications of Artificial Intelligence*, 176, 114741. doi: 10.1016/j.engappai.2026.114741
2. **Wang, H.**, Salunkhe, O., Quadrini, W., Lämkull, D., Ore, F., Despeisse, M., Fumagalli, L., Stahre, J., & Johansson, B. (2024). A systematic literature review of computer vision applications in robotized wire harness assembly. *Advanced Engineering Informatics*, *62*, 102596. doi: 10.1016/j.aei.2024.102596
3. Johansson, B., Despeisse, M., Bokrantz, J., Braun, G., Cao, H., Chari, A., Fang, Q., Chávez, C.A.G., Skoogh, A., Söderlund, H., **Wang, H.**, Wärmefjord, K., Nyborg, L., Sun, J., Örtengren, R., Schumacher, K. A., Espinal, L., Morris, K. C., Nunley, J., Kishita, Y., Umeda, Y., Acerbi, F., Pinzone, M., Persson, H., Charpentier, S., Edström, K., Brandell, D., Gopalakrishnan, M., Rahnama, H., Abrahamsson, L., Rönnbäck, A. Ö., & Stahre, J. (2024). Challenges and opportunities to advance manufacturing research for sustainable battery life cycles. *Frontiers in Manufacturing Technology*, *4*, 1360076. doi: 10.3389/fmtec.2024.1360076
4. Zhu, X., Yu, C., Huang, D., Lei, Z., **Wang, H.**, & Li, S. Z. (2023). Beyond 3dmm: Learning to capture high-fidelity 3d face shape. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *45*(2), 1442-1457. doi: 10.1109/TPAMI.2022.3164131

会议

1. Salunkhe, O., Chávez, C.A.G., **Wang, H.**, Syberfeldt, A., Romero, D., & Stahre, J. (2025). Developing code agents for robot programming: Technical and managerial perspectives. *Advances in Production Management Systems. Cyber-Physical-Human Production Systems: Human-AI Collaboration and Beyond*, 134–147. doi: 10.1007/978-3-032-03515-8_10
2. **Wang, H.**, Salunkhe, O., Quadrini, W., Lämkull, D., Ore, F., Johansson, B., & Stahre, J. (2023). Overview of computer vision techniques in robotized wire harness assembly: Current state and future opportunities. *Procedia CIRP*, *120*, 1071-1076. doi: 10.1016/j.procir.2023.09.127
3. Salunkhe, O., Quadrini, W., **Wang, H.**, Stahre, J., Romero, D., Fumagalli, L., & Lämkull, D. (2023). Review of current status and future directions for collaborative and semi-automated automotive wire harnesses assembly. *Procedia CIRP*, *120*, 696-701. doi: 10.1016/j.procir.2023.09.061
4. Despeisse, M., Johansson, B., Bokrantz, J., Braun, G., Chari, A., Chen, X., Fang, Q., Chávez, C.A.G., Skoogh, A., Stahre, J., Theradapuzha Mathew, N., Turanoglu Bekar, E., **Wang, H.**, & Örtengren, R. (2023). Battery production systems: State of the art and future developments. *Advances in Production Management Systems. Production Management Systems for Responsible Manufacturing, Service, and Logistics Futures*, 521-535. doi: 10.1007/978-3-031-43688-8_36
5. **Wang, H.**, & Johansson, B. (2023). Deep learning-based connector detection for robotized assembly of automotive wire harnesses. *2023 IEEE 19th International Conference on Automation Science and Engineering (CASE)*, 1-8. doi: 10.1109/CASE56687.2023.10260619
6. Zhu, X., **Wang, H.**, Fei, H., Lei, Z., & Li, S. Z. (2021). Face forgery detection by 3d decomposition. *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR),* 2928-2938. doi: 10.1109/CVPR46437.2021.00295
7. Zhu, X., Yang, F., Huang, D., Yu, C., **Wang, H.**, Guo, J., Lei, Z., & Li, S. Z. (2020). Beyond 3dmm space: Towards fine-grained 3d face reconstruction. *Computer Vision - ECCV 2020*, 343-358. doi: 10.1007/978-3-030-58598-3_21

报告

1. 2024 World manufacturing report - New perspectives for the future of manufacturing: Outlook 2030. Available: https://worldmanufacturing.org/report/report-2024-new-perspectives-for-the-future-of-manufacturing-outlook-2030/
2. 2023 World manufacturing report - New business models for the manufacturing of the future. Available: https://worldmanufacturing.org/report/report-2023-new-business-models-for-the-manufacturing-of-the-future/

学位论文

1. **Wang, H.** (2026). Computer vision for non-rigid object assembly automation - With applications in automotive wire harness assembly [PhD thesis].

2. **Wang, H.** (2024). *Toward enabling robotic visual perception for assembly tasks - An application in wire harness assembly onto electric vehicles* [Licentiate thesis]. Available: https://research.chalmers.se/en/publication/540720

科研项目

1. Battery production, products and systems (MAXBATT), VGR, Sweden, 2025-2028
2. Code Agents: AI-powered end-to-end solutions for flexible manufacturing, VINNOVA, Sweden, 2024-2025
3. Boosting the Exploitation of Standardisation Inputs from European Projects (STAND4EU), Horizon Europe, EU, 2022-2024
4. PLENary multi-User developMent arena for future industrial workspaces (PLENUM), VINNOVA, Sweden, 2022-2025
5. DIGITAL work InStructions for cognitive work (DIGITALIS), SIP Produktion2030, VINNOVA, Sweden, 2022-2024
6. Empowering Human Workers for Assembly of Wire Harnesses (EWASS), SIP Produktion2030, VINNOVA, Sweden, 2022-2025
7. A Pan-European Network of Robotics DIHs for Agile Production (DIH^2), Horizon 2020, EU, 2019-2023

学术服务

学术期刊审稿

CMC-Computers, Materials & Continua
International Journal of Industrial Engineering and Management
Neural Processing Letters
Pattern Recognition
Scientific Reports
Signal, Image and Video Processing

学术会议审稿

IEEE International Conference on Multimedia and Expo (ICME)
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

教学经历

查尔姆斯理工大学

Automation Technique (LMT108)
Computer Aided Design (PPU156)
Production Systems (PPU161)
Research Methodology in Production Projects (PPU215)
Robotics and Manufacturing Automation (MPR213)
Simulation and Optimisation of Sustainable Production Systems (IMS085)
Simulation and Visualization of Production Systems (IMS020)
Virtual Production (PPU055)

个人荣誉

国家优秀自费留学生奖学金

2024年度国家优秀自费留学生奖学金A类优秀奖

证书资质

雅思7.5，托福105，GRE 325